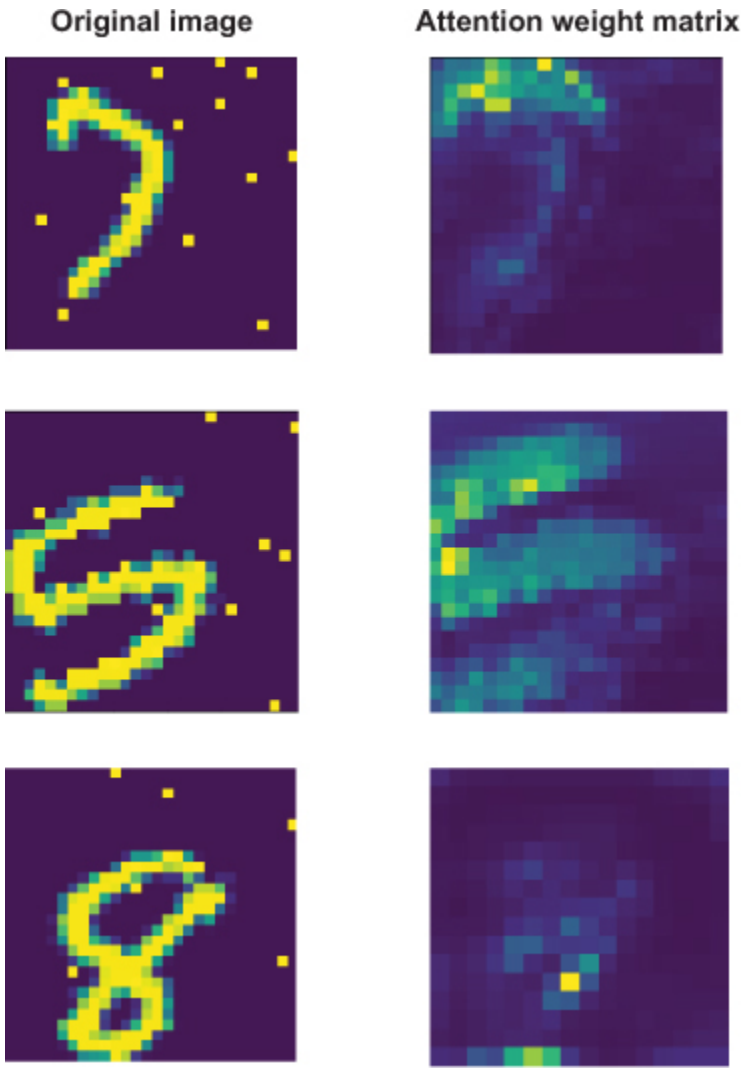
Figure 10.17. Left column: Original input MNIST images (after transformation). Right column: Corresponding self-attention weights showing where the model is paying the most attention.



We visualize this attention map by just reshaping the attention map into a square image:

```
>>> plt.imshow(agent.att_map[0].max(dim=0)[0].view(16,16))
```

The attention weight matrix is a *batch* × *n* × *n* matrix where *n* is the number of nodes, which is $16^2 = 256$ in our example, since after the convolutional layers the spatial extent is reduced from the original 28 × 28. Notice in the top two examples of figure 10.17 that attention maps highlight the contour of the digit but with more intensity at certain parts. If you look through a